

SQL Joins & Aggregations

20 Interview Questions — Northwind Database

The Complete DSA + SQL Interview Preparation Series

Book 5 of 13

For placement preparation, coding interviews, and product-based company rounds.

About This Book

This book is the second SQL installment in the series, focused entirely on joins and aggregations — the two skills that show up in almost every real interview query. You'll practice combining Customers, Orders, OrderDetails, Products, Categories, Employees, Shippers, and Suppliers in different combinations, and summarizing the results with GROUP BY, HAVING, and aggregate functions.

Each question includes a business scenario, the tables involved, the columns your query should return, a sample of what the result might look like, and the core skills being tested. As always, no solution query is provided — writing it yourself is the point.

How To Use This Book

For every question, first sketch the join path on paper: which tables need to be connected, and on which keys. Then decide what needs to be grouped and what needs to be filtered before versus after grouping (WHERE vs. HAVING). Use the sample result only to sanity-check the shape of your output, not as something to copy.

Difficulty Breakdown

Difficulty	Count
Easy	2
Medium	15
Hard	3

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Q1. Orders With Customer Names

Easy

Topic: INNER JOIN

Business Scenario

A support dashboard needs every order shown alongside the customer's company name instead of just a numeric customer ID, so agents don't have to look up IDs manually.

TABLES USED

Orders, Customers

EXPECTED OUTPUT COLUMNS

OrderID, CompanyName, OrderDate

Sample Result (illustrative)

OrderID	CompanyName	OrderDate
10248	Vins et alcools Chevalier	1996-07-04
10249	Toms Spezialitäten	1996-07-05

Skills Required

INNER JOIN

SELECT

ORDER BY

Interview Tip

Interviewers often check whether you join before or after aggregating — joining after a GROUP BY can silently multiply your totals.

Q2. Full Order Line Items With Product Names

Easy

Topic: INNER JOIN

Business Scenario

A warehouse picking system needs each order line item shown with the actual product name rather than a product ID, along with quantity and unit price, to generate pick lists.

TABLES USED

OrderDetails, Products

EXPECTED OUTPUT COLUMNS

OrderID, ProductName, Quantity, UnitPrice

Sample Result (illustrative)

OrderID	ProductName	Quantity	UnitPrice
10248	Queso Cabrales	12	14.00
10248	Singaporean Hokkien Fried Mee	10	9.80

Skills Required

INNER JOIN

SELECT

Interview Tip

Say your join plan out loud before writing SQL. It shows the interviewer you understand the schema, not just the syntax.

Q3. Total Revenue per Order

Medium

Topic: JOIN / SUM / GROUP BY

Business Scenario

Finance wants the total revenue for every order — quantity multiplied by unit price, summed across all line items in that order, including any discount applied.

TABLES USED

Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

OrderID, TotalRevenue

Sample Result (illustrative)

OrderID	TotalRevenue
10248	440.00
10249	1863.40

Skills Required

JOIN

SUM()

GROUP BY

Interview Tip

Watch for fan-out: joining a one-to-many relationship before SUM() can inflate results if you're not careful about the grain.

Q4. Number of Orders per Customer

Medium

Topic: JOIN / COUNT / GROUP BY

Business Scenario

The account management team wants to rank customers by how many total orders they've placed, to identify the most engaged accounts for a loyalty program.

TABLES USED

Customers, Orders

EXPECTED OUTPUT COLUMNS

CompanyName, OrderCount

Sample Result (illustrative)

CompanyName	OrderCount
Save-a-lot Markets	31
Ernst Handel	30

Skills Required

JOIN	COUNT()	GROUP BY	ORDER BY
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Interview Tip

When a question mentions 'total' or 'combined', ask yourself whether that means SUM, COUNT, or COUNT DISTINCT — they're easy to mix up under pressure.

Q5. Customers With More Than 5 Orders

Medium

Topic: HAVING

Business Scenario

Marketing wants to target only the most frequent customers — those who have placed more than 5 orders in total — for an exclusive early-access campaign.

TABLES USED

Customers, Orders

EXPECTED OUTPUT COLUMNS

CompanyName, OrderCount

Sample Result (illustrative)

CompanyName	OrderCount
Ernst Handel	30
QUICK-Stop	28

Skills Required

JOIN	GROUP BY	HAVING	COUNT()
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Interview Tip

HAVING filters groups after aggregation; WHERE filters rows before. Mixing these up is one of the most common interview mistakes.

Q6. Revenue by Product Category

Medium

Topic: Multi-table JOIN / SUM

Business Scenario

The category management team wants total revenue generated by each product category, to decide which categories deserve more shelf space and marketing budget next quarter.

TABLES USED

Categories, Products, OrderDetails

EXPECTED OUTPUT COLUMNS

CategoryName, TotalRevenue

Sample Result (illustrative)

CategoryName	TotalRevenue
Beverages	286526.95
Dairy Products	251330.50

Skills Required

JOIN	SUM()	GROUP BY	ORDER BY
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Interview Tip

For revenue-style questions, always clarify whether discounts should be applied before or after computing the total.

Q7. Employees and Their Total Sales

Medium

Topic: JOIN / SUM / GROUP BY

Business Scenario

Sales leadership wants a leaderboard of every employee's total sales value (based on the orders they processed), sorted from highest to lowest, for a quarterly review.

TABLES USED

Employees, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

FirstName, LastName, TotalSales

Sample Result (illustrative)

FirstName	LastName	TotalSales
Margaret	Peacock	232890.85
Janet	Leverling	202812.75

Skills Required

JOIN	SUM()	GROUP BY	ORDER BY
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Interview Tip

Ranking questions (top N, most/least) almost always need an ORDER BY paired with a LIMIT or equivalent.

Q8. Products Supplied by Each Supplier

Easy

Topic: JOIN / COUNT

Business Scenario

Procurement wants to know how many different products each supplier currently provides, to identify suppliers the company is most dependent on.

TABLES USED

Suppliers, Products

EXPECTED OUTPUT COLUMNS

CompanyName, ProductCount

Sample Result (illustrative)

CompanyName	ProductCount
Exotic Liquids	3
New Orleans Cajun Delights	4

Skills Required

JOIN	COUNT()	GROUP BY
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Interview Tip

For supplier or category counts, decide up front whether you need COUNT(*) or COUNT(DISTINCT column) — they are not interchangeable.

Q9. Average Discount Given per Category

Medium

Topic: JOIN / AVG / GROUP BY

Business Scenario

Finance is investigating margin erosion and wants the average discount applied across all order line items, broken down by product category.

TABLES USED

Categories, Products, OrderDetails

EXPECTED OUTPUT COLUMNS

CategoryName, AverageDiscount

Sample Result (illustrative)

CategoryName	AverageDiscount
Confections	0.06
Seafood	0.05

Skills Required

JOIN

AVG()

GROUP BY

Interview Tip

Average-based questions can hide edge cases: what happens if a group has zero rows? Think about whether AVG() would return NULL.

Q10. Customers Who Have Never Ordered

Medium

Topic: LEFT JOIN / NULL Check

Business Scenario

A re-engagement campaign needs a list of every customer in the system who has never placed a single order, so the marketing team can send them a first-purchase incentive.

TABLES USED

Customers, Orders

EXPECTED OUTPUT COLUMNS

CompanyName, City

Sample Result (illustrative)

CompanyName	City
FISSA Fabrica Inter. Salchichas	Madrid

Skills Required

LEFT JOIN

IS NULL

Interview Tip

LEFT JOIN plus IS NULL is the standard pattern for 'entities that never matched' questions — memorize this shape.

Q11. Shippers Ranked by Number of Orders Shipped

Medium

Topic: JOIN / COUNT / GROUP BY

Business Scenario

Logistics wants to compare shipping partners by how many orders each has handled, to renegotiate rates with the most-used carrier.

TABLES USED

Shippers, Orders

EXPECTED OUTPUT COLUMNS

CompanyName, OrdersShipped

Sample Result (illustrative)

CompanyName	OrdersShipped
United Package	326
Federal Shipping	255

Skills Required

JOIN	COUNT()	GROUP BY	ORDER BY
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Interview Tip

For carrier or partner comparisons, confirm whether cancelled or unfulfilled orders should be excluded from the count.

Q12. Top 3 Best-Selling Products by Quantity

Medium

Topic: JOIN / SUM / LIMIT

Business Scenario

The merchandising team wants the three products with the highest total quantity sold across all orders, to decide what to feature on the homepage.

TABLES USED

Products, OrderDetails

EXPECTED OUTPUT COLUMNS

ProductName, TotalQuantitySold

Sample Result (illustrative)

ProductName	TotalQuantitySold
Camembert Pierrot	1577
Raclette Courdavault	1496

Skills Required

JOIN	SUM()	GROUP BY	ORDER BY	LIMIT
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Interview Tip

When a question says 'top N', clarify ties: should exactly N rows come back, or every row tied for Nth place?

Q13. Average Order Size by Country

Medium

Topic: JOIN / AVG / GROUP BY

Business Scenario

Regional strategy wants to compare average order value across customer countries, to prioritize which regions deserve more sales attention next year.

TABLES USED

Customers, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

Country, AverageOrderValue

Sample Result (illustrative)

Country	AverageOrderValue
Austria	1424.60
Venezuela	902.10

Skills Required

JOIN

AVG()

GROUP BY

Interview Tip

Country- or region-level aggregates often hide null or inconsistent values — decide how you'd handle a missing country field.

Q14. Employees Who Report to a Given Manager

Medium

Topic: Self Join

Business Scenario

HR wants an org-chart export showing which employees report directly to a specific manager, using the employee table's own self-referencing ReportsTo column.

TABLES USED

Employees (self join)

EXPECTED OUTPUT COLUMNS

EmployeeName, ManagerName

Sample Result (illustrative)

EmployeeName	ManagerName
Janet Leverling	Andrew Fuller
Margaret Peacock	Andrew Fuller

Skills Required

Self Join

INNER JOIN

Interview Tip

Self joins are easiest to reason about if you mentally rename the two copies of the table (e.g., e1 for employee, e2 for manager).

Q15. Categories With No Discontinued Products

Medium

Topic: Subquery / NOT IN

Business Scenario

Product management wants to know which categories currently have zero discontinued products, meaning every item in that category is still actively sold.

TABLES USED

Categories, Products

EXPECTED OUTPUT COLUMNS

CategoryName

Sample Result (illustrative)

CategoryName
Beverages
Grains/Cereals

Skills Required

Subquery

NOT IN

GROUP BY

Interview Tip

'Has none of X' questions (categories with no discontinued products) are a strong signal to reach for NOT IN or NOT EXISTS.

Q16. Orders Shipped Later Than 30 Days After Order Date

Medium

Topic: JOIN / Date Math

Business Scenario

Operations wants to flag every order where the gap between OrderDate and ShippedDate exceeded 30 days, to investigate recurring fulfillment delays.

TABLES USED

Orders

EXPECTED OUTPUT COLUMNS

OrderID, OrderDate, ShippedDate, DaysToShip

Sample Result (illustrative)

OrderID	OrderDate	ShippedDate	DaysToShip
10265	1996-07-25	1996-08-12	18
10287	1996-08-19	1996-08-28	9

Skills Required

Date Functions	WHERE	Calculated Columns
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Interview Tip

Date-difference questions depend heavily on the database engine's date functions — be ready to explain the logic even if exact syntax varies.

Q17. Suppliers From Countries With More Than 2 Suppliers

Medium

Topic: Subquery / HAVING

Business Scenario

Procurement wants to identify supplier-dense countries — those contributing more than two suppliers — to assess geographic concentration risk in the supply chain.

TABLES USED

Suppliers

EXPECTED OUTPUT COLUMNS

Country, SupplierCount

Sample Result (illustrative)

Country	SupplierCount
USA	4
UK	3

Skills Required

GROUP BY

HAVING

COUNT()

Interview Tip

Whenever a question involves 'more than N of something', HAVING with a COUNT() is almost always the mechanism.

Q18. Products Priced Above Their Category Average

Hard

Topic: Correlated Subquery

Business Scenario

Pricing strategy wants to flag every product priced above the average price of its own category, to review whether the premium is justified.

TABLES USED

Products, Categories

EXPECTED OUTPUT COLUMNS

ProductName, CategoryName, UnitPrice

Sample Result (illustrative)

ProductName	CategoryName	UnitPrice
Côte de Blaye	Beverages	263.50
Thüringer Rostbratwurst	Meat/Poultry	123.79

Skills Required

Correlated Subquery	AVG()	JOIN
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Interview Tip

Correlated subqueries (like 'above the category average') run once per outer row — be ready to explain why that's more expensive than a plain aggregate.

Q19. Monthly Revenue Trend

Hard

Topic: Date Functions / GROUP BY

Business Scenario

Leadership wants a month-by-month revenue trend across all orders for the year, to visualize seasonality in a board presentation.

TABLES USED

Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

OrderMonth, TotalRevenue

Sample Result (illustrative)

OrderMonth	TotalRevenue
1996-07	27861.90
1996-08	25485.28

Skills Required

Date Functions	GROUP BY	SUM()
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Interview Tip

Trend-over-time questions usually want you to truncate or format a date into a period (month, quarter) before grouping.

Q20. Customers Whose Total Spend Exceeds the Overall Average

Hard

Topic: Subquery / Aggregation

Business Scenario

Finance wants to identify every customer whose total lifetime spend is above the average total spend across all customers, as a shortlist for a VIP account program.

TABLES USED

Customers, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

CompanyName, TotalSpend

Sample Result (illustrative)

CompanyName	TotalSpend
QUICK-Stop	117483.39
Ernst Handel	104874.98

Skills Required

Subquery	SUM()	AVG()	HAVING
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Interview Tip

'Above the overall average' questions require computing the average first, often in a subquery, before filtering the outer result against it.